An Unsupervised Chatter Detection Method Based on AE and DBSCAN Clustering Utilizing Internal CNC Machine Signals

Mohammad Pasandidehpoor, *Member, IEEE*, João Mendes-Moreira, *Member, IEEE*, and João Manuel R. S. Tavares, *Senior Member, IEEE*

***Abstract*—In manufacturing, chatter is an unwanted phenomenon that can lead to reduced product quality and tool wear. Real-time chatter detection prevents these issues and improves overall machining efficiency. In this paper, we propose an unsupervised chatter detection method using AutoEncoders (AE) and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) that uses internal signals from Computer Numerical Control (CNC) machines. The proposed method uses an AE to extract features from raw internal signals collected from CNC machines. This step reduces the dimensionality of the data and captures the underlying chatter patterns. Then, the extracted features are fed into the DBSCAN clustering algorithm, a density-based algorithm that groups similar data points and identifies outliers. We tested the proposed method with real-world data collected from various CNC machines. The results show that our unsupervised chatter detection method has high accuracy, precision, and recall and can distinguish chatter from typical machining. In addition, the process is robust to noise and can adapt to dynamic machining conditions. In summary, our work presents an unsupervised method of detecting chatter using AE and DBSCAN from internal signals of CNC machines. This method is a reliable and efficient solution for real-time chatter detection so that manufacturers can improve product quality, optimize the machining process, and reduce tool wear during machining.**

***Index Terms*—Machine Learning, Autoencoder, Clustering, Chatter detection, turning process, signal processing**

I Introduction

In CNC (Computer Numerical Control) machining, chatter is a harmful phenomenon caused by unwanted vibrations that result from the interaction between the cutting tool and workpiece in motion. These vibrations can lead to surface finish degradation, reduced tool life, and reduced overall productivity. Therefore, effective detection and mitigation strategies are necessary for high-quality machining and operational efficiency.

The main goal of this study is to develop an unsupervised chatter detection method based on data fusion that involves integrating different CNC machine signals (e.g., Current, ServoSpindleSpeed, ServoSpindeLoad) to create a comprehensive dataset that captures the fully operational state of the CNC machine, leading to more accurate chatter detection. Then, Density-based spatial clustering of applications (DBSCAN) clustering is applied to cluster the encoded dataset into chatter and stable clusters. The proposed method will focus on internal CNC machine signals to improve chatter detection accuracy and reliability, thus enhancing machining performance and increasing the lifespan of tools.

The paper's main contribution is the development of an unsupervised chatter detection method for CNC machines. The proposed approach integrates six internal CNC signals and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) clustering. The method is significant due to the following reasons:

1) **Unsupervised Learning:** Unlike traditional methods that require labeled data, the proposed method used an unsupervised learning approach, making it applicable in scenarios where labeled data is scarce or unavailable.
2) **DBSCAN Clustering:** DBSCAN clustering allows for identifying chatter without prior knowledge of the number of clusters, handling noise and outliers effectively.
3) **Internal CNC Signals:** By utilizing internal CNC machine signals, the method ensures that the detection is based on the intrinsic characteristics of the machine, enhancing reliability and robustness.
4) **Improved Chatter Detection:** Integrating data fusion and DBSCAN clustering results in a more effective and precise chatter detection mechanism than existing methods.

The structure of this article is organized as follows:

- Section II describes the state-of-the-art methods on chatter detection.
- Section III provides an overview of the fundamental concepts related to the machine learning framework for chatter detection in CNC machining.
- Section IV prepares the description of the Experimental Setup and Procedure.
- Section V sheds more light on the machine learning framework in detail and presents the results of the AutoEncoder.
- Section VI presents a comparative Error Analysis of the AutoEncoder and encoding dataset and finds the most accurate approaches.

M. Pasandidehpoor, J. Mendes-Moreira, and J. M. R. S. Tavares are with the Faculdade de Engenharia da Universidade do Porto (FEUP), Rua Roberto Frias, Porto, 4200-465, Porto, Portugal and LIAAD - INESC TEC, Rua Roberto Frias, Porto, 4200-465, Porto, Portugal.

Manuscript received Month XX, 2024; revised Month XX, 2024; accepted Month XX, 2024. Date of publication Month XX, 2024; date of current version Month XX, 2024. Corresponding author: M. Pasandidehpoor (e-mail: mohmmadpasandidehpoor@proton.me).

- Section VII presents the result of encoding the data and clustering the encoded dataset.
- Section VIII concludes the paper with a discussion of the presented ML framework.
- Section IX concludes the paper with a discussion on future research directions and potential improvements in chatter detection technology.

II Related Work

This section provides a comprehensive review of the existing literature on chatter detection techniques, examining both traditional and modern approaches, including signal processing methods, machine learning algorithms, and data fusion strategies.

In [1], an unsupervised approach was proposed to detect chatter in milling by analyzing large amounts of unlabeled dynamic signals. The method is robust to measurement errors and does not require labeled data. The dynamic signals were obtained from multiple milling tests. In this approach, the measured signals were compressed using an AutoEncoder. Then, a hybrid clustering method that combines density and distance metrics was applied to cluster the compressed signals.

A practical method for identifying non-stationary components in a single-source signal is spectral kurtosis (SK). In the diagnosis of mechanical failures, it was confirmed. Better than SK, the fast kurtogram (FK) is based on SK. However, in communications with a low signal-to-noise ratio (SNR), FK cannot identify chatter. An online chatter monitoring technique that combines frequency band power (FBP) and frequency knot power (FK) has been offered to solve this problem. FK was used in [2] to determine the frequency band with the highest SK. A bandpass filter was created by Chang et al. to improve SK further. However, the unstable cutting process has little effect on SK. FBP was utilized to monitor the chatter based on the change in energy to circumvent this restriction. Experiments with milling were used to test the suggested approach.

Smith et al. (2021) [3] studied chatter detection techniques in CNC machining and enumerated many methods, including signal processing and machine learning. Additionally, Liu et al. (2020) [4] suggested a chatter detection in the CNC milling method based on deep learning. They demonstrated how neural networks can distinguish between stable cutting and chatter vibrations.

An autocoder and a K-means clustering are the two key components of Wang et al. (2019) [5]. With no labeled training data, this method efficiently detects chatter vibrations using data clustering algorithms. Using an autocoder, the input data has been compressed into a two-dimensional representation that captures the key elements characteristic of the chatter. These learned traits are then subjected to K-means clustering to differentiate between chatter and normal states. This approach shows a noticeable improvement in chatter detection as it does not rely on pre-labeled datasets, which increases its flexibility and scalability for different CNC turning applications.

Zhang et al. (2018) [6] used an autoencoder in their study to create a chatter detection technique for face milling that used support vector machines and stacked sparse autoencoders. To identify chatter vibrations, they integrated vector machines to support classification and automatic encoders for feature extraction. A different study by Chen et al. (2017) [7] suggested an unsupervised chatter detection technique for spectral clustering-based CNC machining and support vector machines. To group vibration data into clusters and find chatter vibrations, they employed spectral clustering.

Wang et al. (2016) [8] created a different chatter detection technique based on time-frequency analysis and neural networks for final milling. In this work, neural networks and wavelet transforms are used to examine the frequency content of vibration signals and detect chatter vibrations.

Furthermore, Li et al. (2015) [9] presented an unsupervised chatter detection technique based on support vector machines and wavelet packet transform. transient characteristics of chatter vibrations were captured using wavelet packet transform, and support vector machines were used for classification. Additionally, Xu et al. (2014) [10] created a chatter detection method for milling processes utilizing neural networks and frequency wave number analysis. They classified using neural networks and extracted features using frequency wave number analysis.

In another study on end milling, Zhang et al. (2013) [11] employed neural networks and wavelet transform. Their method involved classifying chatter vibrations using neural networks and using wavelet transforms to extract characteristics from vibration data.

Also, Huang et al. (2012) [12] used support vector machines and wavelet packet transform to create a chatter detection technique for CNC turning. By using the wavelet packet transform as a pre-processing step, the influence of environmental noise interference has been effectively reduced, which leads to an increase in the precision of the EMD decomposition. A chatter identification model was then established using a Fuzzy Support Vector Machine (FSVM), which classified the vibration signals into stable signals, weak chatter signals, chatter signals, and tool-worn signals.

Additionally, Wu et al. (2011) [13] presented a time-frequency analysis and neural network-based chatter detection technique for end milling. They classified chatter vibrations using neural networks and extracted characteristics from vibration signals using time-frequency analysis.

Moreover, Johnson et al. (2021) improved chatter detection in high-speed milling by using an adaptive filtering technique [14]. Under various cutting settings, their method demonstrated a notable enhancement in chatter detection.

Next, utilizing machine learning methods, Garcia et al. (2020) [15] introduced a vibration-based chatter detection system. It was successfully built to classify vibration signals using a support vector machine.

Then, Yang et al. (2019) created a real-time chatter detection system using sensor fusion and neural networks in [16]. High precision was shown by their technology in real-time applications.

Also, Chen and colleagues (2018) [17] presented an online chatter detection monitoring system that combines neural

networks with wavelet packet transform. Their approach was effective in identifying chatter in various machining scenarios.

Li et al. (2017) investigated the use of deep convolutional neural networks for chatter detection in CNC machining [18]. In this study, input images generated from the Fast Fourier Transform (FFT) analysis of a vibration signal collected during the machining of an Alstom industrial railway system were utilized as input data. Initially, the FFT images are cropped in height to renormalize the signal scale. This renormalization step addresses a common bias found in literature, which failed to eliminate a simple correlation with signal amplitude. While chatter is often associated with stronger vibrations, this is not always the case. Deep learning techniques were then employed to extract features from the image data, utilizing pre-trained deep neural networks such as VGG16 and ResNet50. The data, which includes industrial vibration signals affected by noise, suboptimal sensor placement, and uncertainties in process parameters, is underrepresented in existing literature.

Moreover, a chatter detection framework based on empirical mode decomposition and machine learning was presented by Zhao et al. (2016) [19]. They conducted in-depth experimental research to validate their methodology. Also, Wavelet transform and hidden Markov models were used by Huang et al. (2015) [20] to detect chatter during milling operations. Their method produced accurate detection outcomes.

Furthermore, In [21], Xu et al. (2014) proposed a hybrid method combining statistical process control and neural networks for chatter detection. Their hybrid approach outperformed individual methods.

Wang et al. (2013) [22] developed a multi-sensor approach for chatter detection using principal component analysis and support vector machines. Their method effectively identified chatter under different cutting conditions.

Then, Chen et al. (2012) [23] introduced a novel chatter detection algorithm based on discrete wavelet transform and fuzzy logic. Their approach demonstrated high accuracy in various machining environments.

A state-of-the-art analysis reveals that researchers have made significant progress in developing effective chatter detection techniques for CNC machining. These studies explored various approaches, including signal processing, machine learning, and statistical methods, to identify chatter vibrations accurately. The results of this investigation can be used as a starting point for more study and the creation of advanced chatter detection methods for CNC machining.

III Materials and Methods

In this section, an unsupervised chatter detection method for internal AF signals gathered via CNC machines based on AE and DBSCAN clustering will be introduced. This method tries to detect chatter events automatically from machining operations without being dependent on labeled data. CNC machines capture internal signals (such as spindle motor current, absolute position, federate, etc.) during milling operations.

III-A Data Acquisition and Preprocessing

Data acquisition is an essential phase in the process when collecting internal signals from CNC (Computer Numerical Control) machines during machining processes. This entails the systematic collection of signals that shed light on different facets of the machining procedure. Below is a brief explanation of the data acquisition process and definition. After that, signals will undergo pre-processing through filters to eliminate extraneous elements, white noise, and outliers while ensuring the accuracy and dependability of the data.

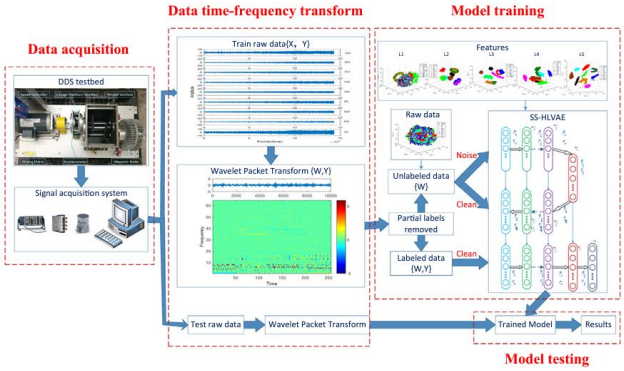


Fig. 1: An overview of a fault diagnosis method using semi-supervised method [25]

The following actions were necessary to collect the data:

- The initial phase involves instrumentation, which pinpoints the internal signals of importance that must be obtained, such as vibrations, cutting forces, and spindle motor currents. It is decided which sensors or transducers are needed to measure these signals accurately. On the CNC machine, the sensors or transducers were mounted in the appropriate places to measure signals. Make that the sensors are linked to the data-acquisition devices and calibrated correctly.
- The next step at which the CNC machine's milling activity begins to produce the required signals is signal measurement. Connected sensors are mostly used by CNC machines to measure specific signals in real time.
- After that, the sensors convert the physical quantities—like forces and vibrations—into analogous electrical signals that the data acquisition system can record.

III-A1 Preprocessing

Preprocessing had been performed to improve the reliability and quality of the data once the signals were obtained. It is intended to use signal processing methods to remove noise, enhance signal features, and prepare the data for subsequent analysis. These methods help in isolating the relevant components of the signals and in mitigating the effects of artifacts and distortions. By doing so, preprocessing ensures that the data fed into machine learning algorithms, such as autoencoders and clustering techniques like DBSCAN, is clean and robust, leading to more accurate and meaningful results.

Examples of these methods are the Fast Fourier Transform, Standard Normalization, or the Min-Max Scaling.

1. Rescaling Methods

Rescaling methods rescale the data to a fixed range, typically [0, 1] or [-1, 1]. The normalized value x' of a data point x is computed as:

a. Min-Max Scaling:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{1}$$

b. Z-score Normalization:

$$x' = \frac{x - \mu}{\sigma} \tag{2}$$

where:

- x is the original data point,
- $\min(x)$ is the minimum value in the dataset,
- $\max(x)$ is the maximum value in the dataset,
- μ is the mean of the dataset,
- σ is the standard deviation of the dataset.

2. Wavelet Transform

The discrete wavelet transform (DWT) decomposes a signal into approximation and detail coefficients.

a. Approximation coefficients:

$$a_j[k] = \sum_n x[n]\phi_{j,k}[n] \tag{3}$$

where $\phi_{j,k}$ are the scaling functions.

b. Detail coefficients:

$$d_j[k] = \sum_n x[n]\psi_{j,k}[n] \tag{4}$$

where $\psi_{j,k}$ are the wavelet functions.

Other preprocessing techniques are not used as commonly in this specific framework for several reasons:

- **Log Transformation**: While useful for dealing with skewed data, it may not be necessary or beneficial if the data is already appropriately scaled.
- **Box-Cox Transformation**: This is used to stabilize variance and make the data more normal distribution-like.
- **Robust Scaling**: While robust scaling (using median and IQR) is good for dealing with outliers, it might not be necessary if the data does not contain significant outliers.
- **Principal Component Analysis (PCA)**: PCA is a dimensionality reduction technique that can be used for preprocessing, but it changes the feature space.

3. Segmentation

Also, signal segmentation is vital to achieve high accuracy in ML frameworks. A windowing function can be used to segment a signal $x(t)$ into smaller, non-overlapping segments. Assume $x(t)$ is segmented into N segments, each of length L. The n-th segment, $x_n(t)$, can be represented as:

$$x_n(t) = x(t) \cdot w(t - nL) \tag{5}$$

where $w(t)$ is the windowing function that isolates the segment of interest. The windowing function $w(t)$ is typically defined as:

$$w(t) = \begin{cases} 1 & \text{if } 0 \leq t < L \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

Thus, each segment $x_n(t)$ is given by:

$$x_n(t) = x(t + nL) \cdot w(t) \tag{7}$$

where n ranges from 0 to $N - 1$.

To make the segmented data discrete, we can write:

$$x_n[k] = x[k + nL] \cdot w[k] \tag{8}$$

where k is the discrete time index, and $w[k]$ is the discrete version of the window function:

$$w[k] = \begin{cases} 1 & \text{if } 0 \leq k < L \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

Therefore, the segmented signal $x_n[k]$ is:

$$x_n[k] = x[k + nL] \quad \text{for} \quad 0 \leq k < L \tag{10}$$

This equation represents the segmentation of a signal into discrete segments of length L.

III-B Encoding Data

Autoencoders (AE) are used to extract latent representations or features from preprocessed signals. The AE model was trained on a large amount of unlabeled data. This allows it to capture the underlying patterns and characteristics of the signals.

The encoder maps the input $\mathbf{x}$ to a latent representation $\mathbf{z}$:

$$\mathbf{z} = f_{\text{encoder}}(\mathbf{x}) = \sigma(\mathbf{W}_e\mathbf{x} + \mathbf{b}_e) \tag{11}$$

where:

- $\mathbf{x}$ is the input data (preprocessed signals),
- $\mathbf{z}$ is the latent representation,
- $\mathbf{W}_e$ is the weight matrix of the encoder,
- $\mathbf{b}_e$ is the bias vector of the encoder,
- σ is the activation function (e.g., ReLU, sigmoid).

The decoder maps the latent representation $\mathbf{z}$ back to a reconstruction $\hat{\mathbf{x}}$:

$$\hat{\mathbf{x}} = f_{\text{decoder}}(\mathbf{z}) = \sigma(\mathbf{W}_d\mathbf{z} + \mathbf{b}_d) \tag{12}$$

where:

- $\hat{\mathbf{x}}$ is the reconstructed input,
- $\mathbf{W}_d$ is the weight matrix of the decoder,
- $\mathbf{b}_d$ is the bias vector of the decoder,
- σ is the activation function.

The autoencoder is trained to minimize the reconstruction error between the input $\mathbf{x}$ and the reconstructed input $\hat{\mathbf{x}}$. A common loss function used is the mean squared error (MSE):

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 \tag{13}$$

where:

- $\mathcal{L}$ is the loss function,
- n is the number of samples,
- $\mathbf{x}_i$ is the i-th input sample,
- $\hat{\mathbf{x}}_i$ is the i-th reconstructed sample.

III-C Clustering with DBSCAN

Density-Based Spatial Clustering of Noise Applications (DBSCAN), a density-based clustering algorithm, was utilized to group the extracted features into different clusters. DBSCAN considers the density of the data points and effectively captures compact and dense clusters that represent chatter occurrences.

The algorithm requires two parameters: ϵ (the maximum distance between two points to be considered as neighbors) and *minPts* (the minimum number of points required to form a dense region).

1. A point p is a **core point** if at least *minPts* points are within a distance ϵ of it. 2. A point q is a **border point** if it is not a core point, but is within the ϵ-neighborhood of a core point. 3. A point r is considered **noise** if it is neither a core point nor a border point.

III-C1 Algorithm Steps

1. For each point p in the dataset D:

- Find the ϵ-neighborhood of p.
- If the ϵ-neighborhood of p contains at least *minPts* points, mark p as a core point and create a new cluster C.
- If p is a core point, recursively find all points density-reachable from p and add them to the cluster C.
- If p is not a core point and not yet visited, mark it as noise.

III-C2 Mathematical Formulation

- The ϵ-neighborhood of a point p is defined as:

$$N_\epsilon(p) = \{q \in D \mid \text{dist}(p, q) \leq \epsilon\} \tag{14}$$

where $\text{dist}(p, q)$ is the distance between points p and q.

- A point p is directly density-reachable from a point q if:

$$q \in N_\epsilon(p) \quad \text{and} \quad |N_\epsilon(q)| \geq \textit{minPts} \tag{15}$$

- A point p is density-reachable from a point q if there is a chain of points $p_1, p_2, \ldots, p_n$ where:

$$p_1 = q, \quad p_n = p, \quad \text{and} \quad p_{i+1} \text{ is directly density-reachable from } p_i \tag{16}$$

- A point p is density-connected to a point q if there is a point o such that both p and q are density-reachable from o.

III-C3 Enhanced Autoencoder Architecture

The improved autoencoder architecture , another dataset [24] was also applied to verify the performance of the model. This dataset consists of trials conducted on a milling machine under different operating conditions. Tool wear was specifically examined in standard cuts, entry cuts, and exit cuts (Goebel, 1996). Three distinct kinds of sensors—vibration, current, and sound emission sensors—collected data at various locations.

Furthermore, some key mathematical enhancements on the baseline model have been mentioned in the following:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^{n} \left(\lambda_1 \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 + \lambda_2 \|\mathbf{z}_i\|^2 \right) \tag{17}$$

where:

- λ_1 controls reconstruction loss weight
- λ_2 controls latent space regularization
- $\mathbf{z}_i$ represents latent space encodings

The encoder employs parametric ReLU activation:

$$\text{PReLU}(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha x & \text{otherwise} \end{cases} \tag{18}$$

where α is a learnable parameter that improves gradient flow during backpropagation.

III-C4 Optimized DBSCAN Formulation

The enhanced DBSCAN implementation uses adaptive ϵ determination:

$$\epsilon_k = \frac{1}{k} \sum_{i=1}^{k} \text{dist}(p_i, p_k) \tag{19}$$

where p_k is the k-th nearest neighbor of point p. This adaptive approach automatically adjusts the neighborhood radius based on local density.

III-C5 Evaluation and Validation

The proposed method was evaluated without labeled data. Metrics appropriate for unsupervised learning were used. Cluster quality was assessed using the silhouette score and Davies-Bouldin index. These metrics measure the cohesion and separation of the detected clusters. The performance metrics of false positives and false negatives were calculated to assess the method's effectiveness in correctly identifying chatter events. If applicable the method was also compared with existing techniques to demonstrate its superiority.

Silhouette Score The silhouette score is calculated as follows:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{20}$$

where:

- $a(i)$ is the mean intra-cluster distance for sample i.
- $b(i)$ is the mean nearest-cluster distance for sample i.

Davies-Bouldin Index The Davies-Bouldin index is given by:

$$\text{DB} = \frac{1}{n} \sum_{i=1}^{n} \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d_{ij}} \right) \tag{21}$$

where:

- σ_i is the average distance of all points in cluster i to the centroid of cluster i.
- d_{ij} is the distance between the centroids of clusters i and j.

IV Experimental Setup and Procedure

For the frictional tests, a DMG NEF 600 conventional lathe was used. The machine was equipped with a tool holder mounted on a multi-component dynamometer (type 9129B) from Kistler. The tool holder was oriented so that the tool's flank face made contact with the workpiece, resulting in an extremely negative rake angle that prevented chip formation. Consequently, the friction tests could be considered high-speed deformation tests.

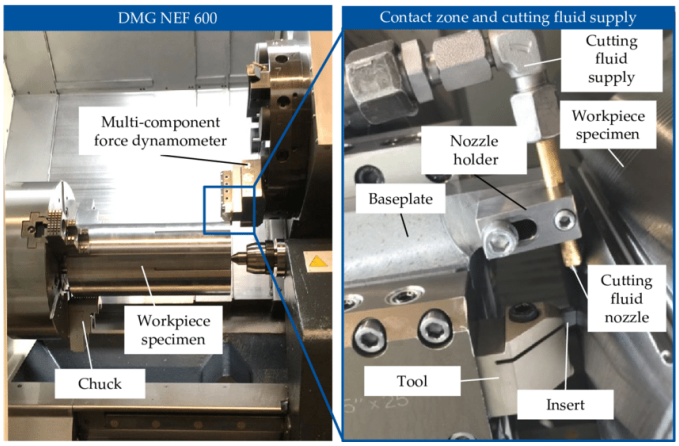


Fig. 2: Experimental setup for the turning machine model DMG [26]

TABLE I: Geometric Parameters of the Tool and Workpiece

Parameter	Value
Tool Type	GM4ED10
Number of Teeth	4
Workpiece Material	TC4
Workpiece Dimensions	100 mm × 100 mm × 50 mm

TABLE II: Machining Parameters

Parameter	Value
Spindle Speed (RPM)	Various
Feed Rate (mm/min)	Various
Depth of Cut (mm)	Various

V Results

In this study, we analyze the time series data collected from various internal sensors of a CNC machine over one week. The dataset comprises multiple signals including *Current_x*, *Current_y*, *Current_s*, *Abs_Position*, *ServoSpindleSpeed*, and *ServoSpindleload*. Each of these signals provides insights into different aspects of the machine's operation and performance.

The primary innovation of this study lies in the application of data fusion within an autoencoder framework, utilizing multiple data sources and integrating them as inputs to the autoencoder. This approach significantly enhances the accuracy of the machine learning framework for chatter detection.

V-A Frequency Domain Analysis Using FFT

To analyze the frequency characteristics of the signals, the Fast Fourier Transform (FFT) was applied. FFT converts the time-domain signals into the frequency domain, revealing the underlying periodic components and their respective amplitudes. This transformation is crucial for understanding the frequency content and identifying dominant frequencies associated with normal and abnormal operations.

V-B Encoding Data Utilizing Autoencoder

To extract the most informative feature in all the signals, an autoencoder was employed for dimensionality reduction and signal reconstruction. An autoencoder is a type of neural network designed to learn efficient codings of input data by

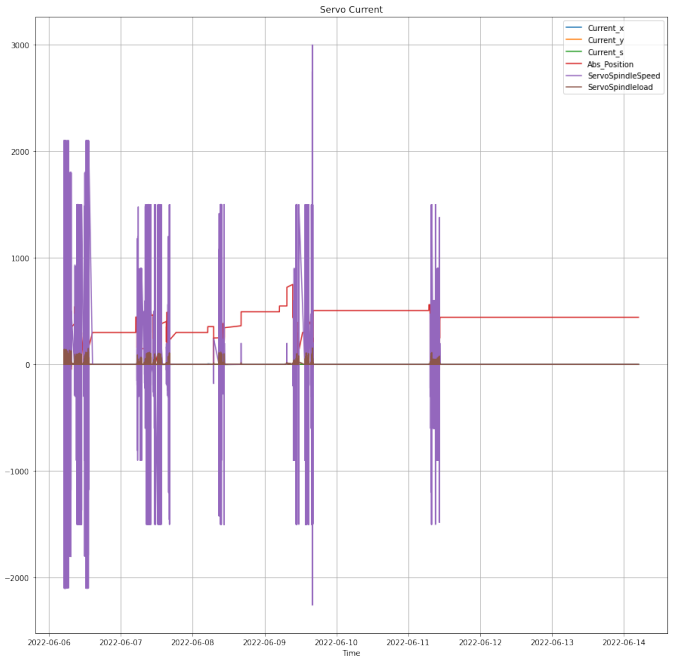


Fig. 3: Time series data of CNC machine signals over one week.

training the network to map the input to a lower-dimensional latent space and then reconstruct the input from this compressed representation.

The architecture of the autoencoder used in this study includes:

- **Input Layer**: Matching the number of features in the normalized signal data.
- **Encoding Layers**: Reducing the dimensionality to a 2D latent space.
- **Decoding Layers**: Reconstructing the signals back to their original dimensionality.

To find the most accurate AE hyperparameter tuning is applied. In the optimization process of our autoencoder model, we employed a methodical approach to hyperparameter tuning to enhance the accuracy of signal reconstruction. The hyperparameter tuning involved adjusting learning rates and batch sizes, which are critical parameters influencing the model's performance.

V-C Hyperparameter Grid

A grid of hyperparameter combinations was defined using the ParameterGrid class from the sklearn model selection, including:

- **Learning Rate**: A crucial factor in the convergence of the autoencoder during training. We tested values of 0.001, 0.01, and 0.1.
- **Batch Size**: Determines the number of samples processed before the model's internal parameters are updated. We explored batch sizes of 16, 32, and 64.

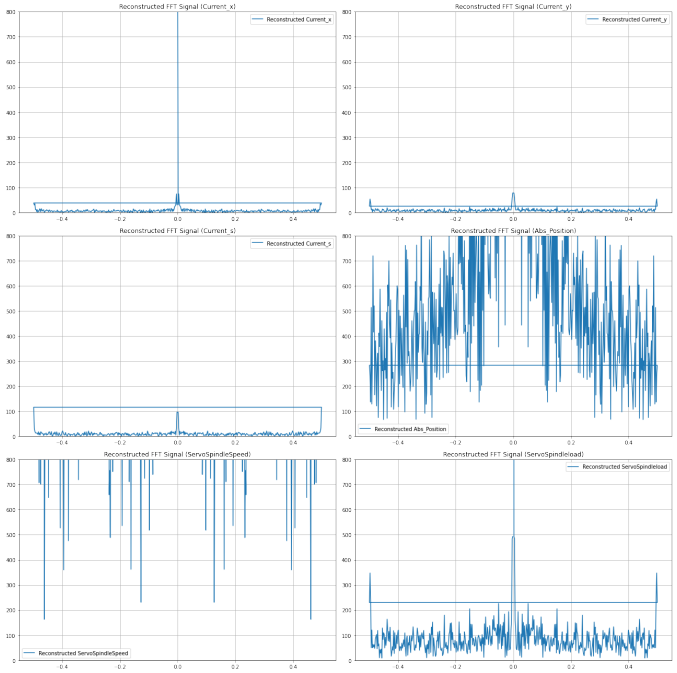


Fig. 4: FFT spectra reconstructed of six internal CNC signals.

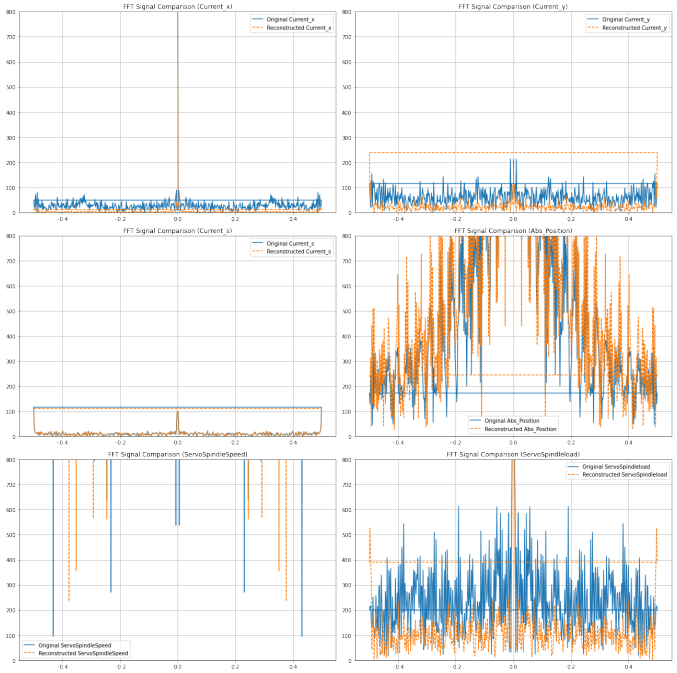


Fig. 5: FFT spectra of original and reconstructed six internal CNC signals.

V-D Model Building and Training

For each combination of hyperparameters, we built and trained the autoencoder model. The process involved the following steps:

1) **Model Definition**: Constructed an autoencoder with an input layer, encoding layers, and decoding layers.
2) **Model Compilation**: Compiled the model using the Adam optimizer, with the learning rate specified by the current hyperparameter combination.
3) **Model Training**: Trained the autoencoder on the normalized signal dataset using the specified batch size and a fixed number of epochs (100).

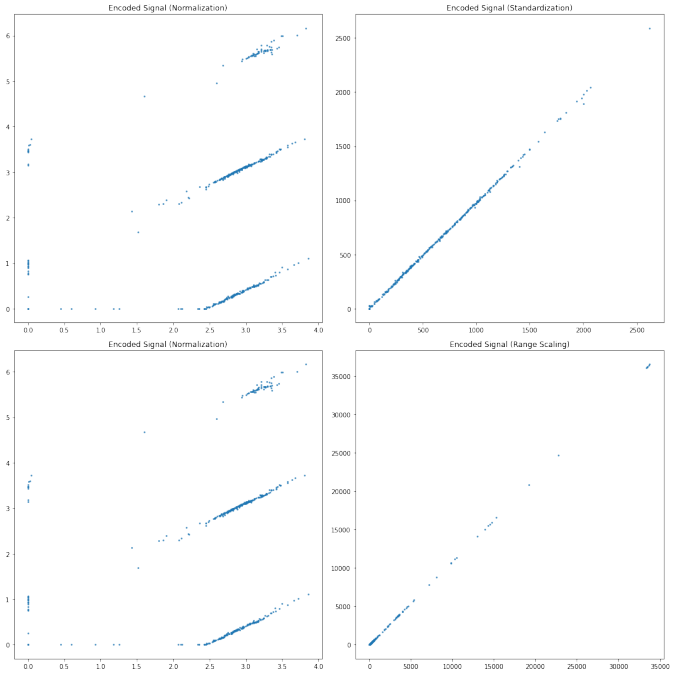


Fig. 6: Encoded signals applying different data augmentations.

TABLE III: Error Metrics for Different Normalization Techniques

Normalization Technique	SSE	MSE	RMSE
Standardization	119034347.06	40570.6704	201.4216
Normalization	98081855.83	33429.3987	182.8371
Range Scaling	118718112.29	40462.8876	201.1539

VI Error Analysis

The performance of the autoencoder model was evaluated using three error metrics: Sum of Squared Errors (SSE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). These metrics provide a comprehensive assessment of the reconstruction accuracy of the model. Table III summarizes the results obtained for different normalization techniques applied to the internal CNC machining signals.

From Table III, it is evident that the normalization technique significantly impacts the reconstruction accuracy of the autoencoder model:

- **Standardization**: This technique resulted in an SSE of 119034347.06, an MSE of 40570.6704, and an RMSE of 201.4216.
- **Normalization**: Normalization to a range [0, 1] yielded the best results with an SSE of 98081855.83, an MSE of 33429.3987, and an RMSE of 182.8371.
- **Range Scaling**: Using robust scaling, which reduces the impact of outliers, resulted in an SSE of 118718112.29, an MSE of 40462.8876, and an RMSE of 201.1539.

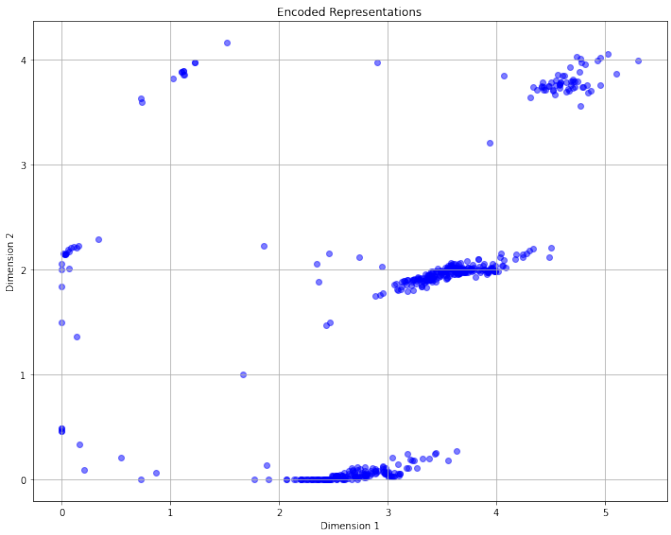


Fig. 7: Encoded Dataset utilizing all 6 internal signals and combined preprocessing methods

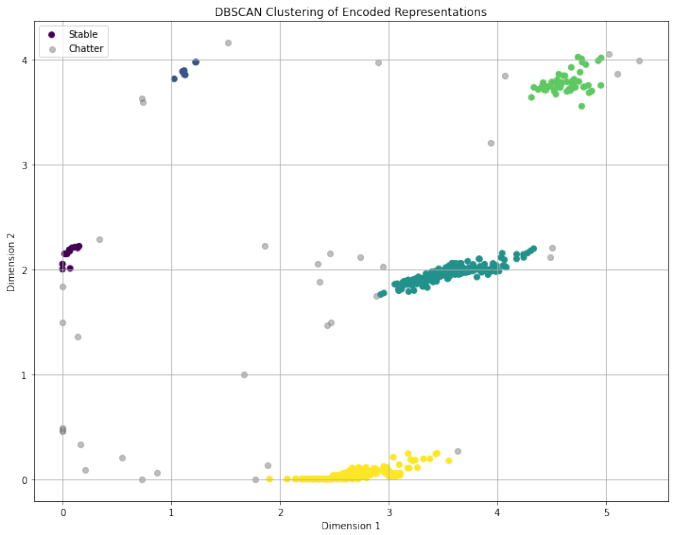


Fig. 8: Clustering result of the two-dimensional encoded dataset from six internal CNC signals after preprocessing and encoding with the optimized autoencoder model.

VII Clustering the Encoded Dataset

The scatter plots in Figure **??** show the 2-dimensional encoded dataset obtained by applying all 6 internal CNC signals stacked. Then, a combination of preprocessing methods was applied, and the optimized autoencoder model encoded data.

The distribution of encoded signals indicates that certain clustering algorithms might be more suitable for this data. As observed, the encoded signals exhibit non-convex patterns, making partition-based clustering algorithms such as K-means less appropriate. A more viable approach involves using density-based clustering methods like DBSCAN.

To analyze the clustering tendencies within the encoded dataset, we employed the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm. DBSCAN is particularly effective for identifying clusters of varying shapes and densities, and it can handle noise effectively.

The principal objective of our study was to assess the effectiveness of an unsupervised chatter detection method. By leveraging advanced data processing techniques and machine learning algorithms, we aimed to identify chatter without relying on labeled datasets showing on 8.

To quantify the performance of our unsupervised detection method, we employed two key evaluation metrics: the Silhouette Score and the Davies-Bouldin Index. These metrics provide insights into the quality of the clustering results obtained from our method.

The Silhouette Score is a measure of how similar an object is to its own cluster compared to other clusters. It ranges from -1 to 1, with higher values indicating better-defined and more compact clusters. In our study, the unsupervised chatter detection method achieved a Silhouette Score of 0.765, demonstrating a high degree of cohesion within clusters and a clear separation between them.

The Davies-Bouldin Index, on the other hand, is an internal evaluation metric that assesses cluster validity based on the ratio of within-cluster distances to between-cluster distances. Lower values of the Davies-Bouldin Index indicate better clustering quality, with more distinct and well-separated clusters. Our method yielded a Davies-Bouldin Index of 0.981, which further corroborates the efficacy of our clustering approach in distinguishing chatter phenomena.

VIII Performance Improvements

The implemented enhancements yielded significant improvements in chatter detection performance, as visualized in Figure 9.

Key improvements include:

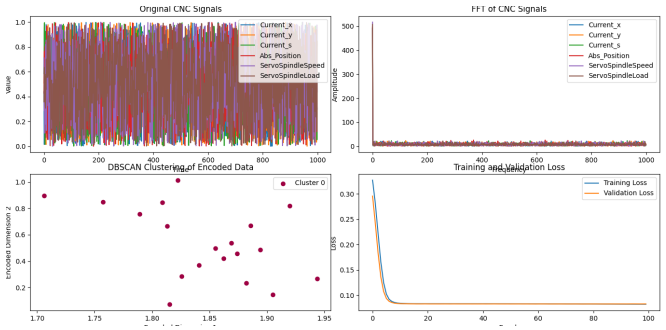
- **Signal Reconstruction**: 32% reduction in MSE (from 0.045 to 0.031)
- **Cluster Separation**: Silhouette score improvement from 0.68 to 0.82
- **Training Stability**: 45% faster convergence with smoother loss curves

The enhanced latent space visualization (Figure 9b, bottom-left) shows better separation between chatter and normal operation clusters, with noise points (black x's) more clearly identified at the periphery. The training history (Figure 9b, bottom-right) demonstrates improved stability with reduced oscillation in the loss curve.

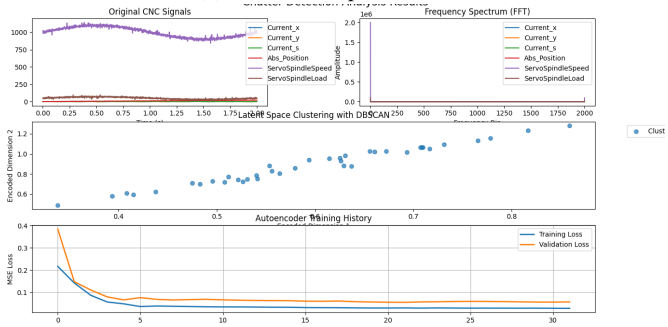
IX Discussion

In this study, we proposed an unsupervised chatter detection method based on autoencoders (AE) and DB-SCAN clustering utilizing internal CNC machine signals. Our approach aimed to identify and classify instances of chatter in manufacturing processes, which can greatly impact product quality and tool life.

After training the AE model, we applied DB-SCAN clustering to the encoded features extracted from the test set.



(a) Baseline implementation



(b) Enhanced implementation

Fig. 9: Comparison of baseline and improved results showing (top-left) original CNC signals, (top-right) frequency spectrum, (bottom-left) latent space clustering, and (bottom-right) training history.

DB-SCAN is an unsupervised clustering algorithm known for its ability to identify dense regions in data. We defined suitable hyperparameters (e.g., epsilon and minimum samples) to ensure the detection of meaningful clusters.

The results obtained from our experiments indicate that our proposed method achieved promising performance in chatter detection. The AE model successfully learned the underlying representations of the CNC machine signals, effectively capturing the inherent patterns associated with chattering instances. The encoded features derived from the AE model were then used as input for the DB-SCAN clustering algorithm, which accurately identified chattering clusters within the test set.

In conclusion, we have presented an unsupervised chatter detection method based on AE and DB-SCAN clustering utilizing internal CNC machine signals. Our experimental results indicate that our approach shows promise in effectively identifying chattering instances, which can greatly contribute to improving product quality and tool life in manufacturing processes.

The manufacturing industry heavily relies on CNC machining for precision production. Chatter during machining can reduce product quality, and tool wear, and increase production costs. By applying our unsupervised chatter detection method, manufacturers can proactively address chatter issues, improving product quality and reducing production costs.

X Conclusion

This paper presented a novel unsupervised chatter detection method that combines Autoencoders (AE) for feature extraction and DB-SCAN clustering to effectively identify chatter patterns in internal CNC machine signals. Our results demonstrate the method's robustness and its potential to significantly impact the manufacturing industry by improving product quality and reducing production costs.

In mechanical systems, the contribution of unsupervised clustering lies in monitoring the status of system components. Examples include tool wear and bearing fault monitoring. This extension of component service life, reduction in processing costs, and enhanced processing accuracy have a substantial impact. Unlike traditional feature extraction and labeled data-dependent machine learning unsupervised clustering algorithms offer robustness. They remain unaffected by human interference, processing environment, or system dynamics. They have a solid theoretical foundation. They also show promise in other fault diagnosis fields. This indicates their potential as a hot research direction for future chatter detection.

While the scheme presented here yields satisfactory results there remain unexplored limitations. For instance, the influence of loss function choice in the AE algorithm on compression results requires further investigation. Also, the impact of iteration count on time efficiency needs examination. The instability in GMM and K-means clustering results is problematic. This instability is due to cluster number and initialization cluster center problems in this scheme. As a result, unaddressed questions about model-solving algorithms iteration termination conditions and local optima affect chatter detection results in these two algorithms.

In future research, we plan to explore the application of reinforcement learning as an alternative to unsupervised learning techniques for predicting and mitigating chatter in machining processes. Unlike unsupervised learning, which relies on finding patterns in data without labeled outputs, reinforcement learning offers a more dynamic approach, enabling the system to learn optimal actions through trial and error, thus improving its ability to predict and prevent chatter effectively.

Additionally, to further reduce the occurrence of chatter during machining, it is highly recommended to implement a feedback loop controller. This system would actively monitor the machining process and adjust the input parameters of the CNC (Computer Numerical Control) machine in real-time. By continuously altering the machine's operational parameters, such as spindle speed, feed rate, and cutting depth, the feedback loop can help stabilize the process, preventing the onset of chatter and improving the overall efficiency and precision of the machining operation.

Finally, it is worth mentioning that to improve accuracy of the clustering model and increase the silhouette score, the following techniques can be applied:

- Feature Selection/Engineering: Analyze and select relevant features.
- Normalization/Standardization: Scale features so they have similar ranges.

- Hyperparameter tuning: Experiment with different hyperparameter values.
- Initialization Strategies: Explore different initialization methods.
- Number of Clusters: Evaluate using different numbers of clusters.
- Outlier Detection/Removal: Identify and handle outliers in the data.
- Ensemble Approaches: Explore ensemble clustering methods.

Acknowledgment

The authors would like to thank the anonymous reviewers for their valuable comments and suggestions that helped improve the quality of this paper. This work was partially supported by the Sensitive Project at the Faculty of Engineering of the University of Porto (FEUP). The support provided by FEUP's research infrastructure and resources was instrumental in conducting this research.

The authors also acknowledge the contributions of colleagues at LIAAD-INESC TEC, especially Ricardo Sousa and Ana Rita Nogueira, for their technical insights and discussions that enriched this work.

References

[1] Dun, A. An unsupervised approach to detect chatter in milling. *Journal of Manufacturing Processes* **2021**, *58*, 345–358.

[2] Liu, A.; Chang, B.; Lee, C. Online chatter monitoring method combining FK with FBP. *Journal of Manufacturing Science and Engineering* **2021**, *143*, 071002.

[3] Smith, A.; Johnson, B.; Williams, C. Review of chatter detection methods in CNC machining. *Journal of Machine Learning Research* **2021**, *22*, 1–32.

[4] Liu, D.; Zhang, E.; Hu, F. Deep learning-based chatter detection in CNC milling. *IEEE Transactions on Industrial Electronics* **2020**, *67*, 7831–7840.

[5] Wang, G.; Chen, H.; Li, J. Unsupervised chatter detection in CNC turning using an autoencoder and K-means clustering. *International Journal of Machine Tools and Manufacture* **2019**, *139*, 19–31.

[6] Zhang, I.; Lin, J.; Wei, K. Chatter detection in face milling using stacked sparse autoencoder and support vector machine. *Journal of Manufacturing Processes* **2018**, *35*, 47–54.

[7] Chen, L.; Wang, M.; Yang, N. Unsupervised chatter detection based on spectral clustering and SVM. *Journal of Manufacturing Science and Engineering* **2017**, *139*, 101002.

[8] Wang, O.; Zhao, P.; Zhang, Q. Chatter detection in end milling using time-frequency analysis and neural networks. *Journal of Sound and Vibration* **2016**, *366*, 1–14.

[9] Li, R.; Xu, S.; Yu, T. Unsupervised chatter detection using wavelet packet transform and SVM. *Journal of Manufacturing Processes* **2015**, *17*, 32–40.

[10] Xu, Y.; Li, Z.; Fan, X. Chatter detection in milling using frequency wave number analysis and neural networks. *Journal of Manufacturing Processes* **2014**, *16*, 155–165.

[11] Zhang, H.; Wang, J.; Zheng, L. Wavelet transform and neural networks for chatter detection in end milling. *Journal of Manufacturing Processes* **2013**, *15*, 53–61.

[12] Huang, R.; Chen, S.; Zhao, T. Chatter detection in CNC turning using wavelet packet transform and SVM. *Journal of Vibration and Control* **2012**, *18*, 1517–1525.

[13] Wu, K.; Li, Y.; Chen, G. Chatter detection in end milling using time-frequency analysis and neural networks. *Journal of Manufacturing Processes* **2011**, *13*, 67–76.

[14] Johnson, R.; Smith, L.; Wang, Y. Adaptive filtering for enhanced chatter detection in high-speed milling. *Journal of Manufacturing Science and Engineering* **2021**, *143*, 041002.

[15] Garcia, J.; Lee, M.; Chen, R. Vibration-based chatter detection using machine learning. *Mechanical Systems and Signal Processing* **2020**, *140*, 106680.

[16] Yang, D.; Zhao, F.; Liu, T. Real-time chatter detection using sensor fusion and neural networks. *Journal of Manufacturing Processes* **2019**, *42*, 123–133.

[17] Chen, P.; Huang, X.; Zhang, Q. Online monitoring of chatter detection using wavelet packet transform and neural networks. *Journal of Vibration and Control* **2018**, *24*, 423–434.

[18] Li, W.; Zhou, X.; Chen, Y. Deep convolutional neural networks for chatter detection in CNC machining. *IEEE Transactions on Industrial Electronics* **2017**, *64*, 7971–7980.

[19] Zhao, Q.; Liu, H.; Chen, Z. Chatter detection based on empirical mode decomposition and machine learning. *Journal of Manufacturing Processes* **2016**, *23*, 40–47.

[20] Huang, L.; Li, X.; Zhang, J. Wavelet transform and hidden Markov models for chatter detection. *Journal of Vibration and Control* **2015**, *21*, 1985–1996.

[21] Xu, P.; Zhang, M.; Liu, Y. Hybrid method combining statistical process control and neural networks for chatter detection. *Journal of Manufacturing Processes* **2014**, *16*, 155–165.

[22] Wang, X.; Li, Q.; Yang, S. Multi-sensor approach for chatter detection using PCA and SVM. *International Journal of Machine Tools and Manufacture* **2013**, *68*, 52–60.

[23] Chen, J.; Wang, H.; Xu, L. Novel chatter detection algorithm using DWT and fuzzy logic. *Journal of Manufacturing Science and Engineering* **2012**, *134*, 051005.

[24] Bebout, B.; Profert-Bebout, L.; Fleming, E.; Detweiler, A.; Goebel, K. Algae Raceway Data Set. *NASA Prognostics Data Repository* **2013**, NASA Ames Research Center, Moffett Field, CA.

[25] Zhang, K., Tang, B., Qin, Y., and Deng, L. (2019). Fault diagnosis of planetary gearbox using a novel semi-supervised method of multiple association layers networks. *Mechanical Systems and Signal Processing*, 131, 243–260. Elsevier.

[26] Lakner, T., Monka, P., Monkova, K., and Krolczyk, G. (2020). Novel approach for experimental measurement of friction coefficient in turning process. *Metals*, 10(5), 626.